

JISHIN BIJUMON GEORGE

ELECTRONICS AND COMMUNICATION ENGINEER | EMBEDDED SYSTEMS & PCB DESIGN ENGINEER

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SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in PCB design, embedded systems, and IoT development. Proficient in KiCad, Altium Designer, ESP32, and Arduino-based system development. Experienced in schematic capture, PCB layout, hardware prototyping, and embedded application development.

INTERNSHIP EXPERIENCE

EVOLVE ROBOTICS, ERNAKULAM

- Intern – Embedded Systems and Robotics
- July 2026 (Ongoing)
- Interfacing embedded systems and robotics projects involving microcontroller programming, sensor interfacing, hardware testing, and prototype development.
- Gaining hands-on experience in debugging electronic systems, integrating hardware modules, and developing practical engineering solutions in a collaborative R&D environment.

NeST DIGITAL , ERNAKULAM

- Intern – IoT and Embedded Systems Immersion Program
- June 2025 (1 Month)
- Worked on IoT and embedded system fundamentals, including microcontroller programming and basic system integration.
- Gained hands-on experience with sensors, embedded applications, and real-time use cases.

AMAL JYOTHI COLLEGE OF ENGINEERING ,(AUTONOMOUS)

KOTTAYAM, KANJIRAPALLY

- Embedded Systems Internship (Arduino IDE)
- May-June 2024 (1 Month)
- Worked on sensor interfacing, Arduino-based embedded C programming, and basic hardware testing for embedded applications.
- Improved understanding of communication protocols, circuit basics, and system-level integration.

EDUCATION

B.TECH IN ELECTRONICS AND COMMUNICATION ENGINEERING

- Amal Jyothi College of Engineering (Autonomous), Kanjirappally
KTU | 2022–2026

HIGHER SECONDARY EDUCATION (CBSE)

- International Indian School, Dammam, Saudi Arabia | 2021–2022

SECONDARY EDUCATION (CBSE)

- International Indian School, Dammam, Saudi Arabia | 2019–2020

TECHNICAL SKILLS:

PCB DESIGN & EDA TOOLS: ALTIUM DESIGNER, KICAD, ORCAD

EMBEDDED PLATFORMS: ESP32, ESP8266, ARDUINO

EMBEDDED & HARDWARE SKILLS: Sensor Interfacing ,PCB Debugging, Hardware Testing

PROGRAMMING & DEVELOPMENT TOOLS: C and Python (basics), ArduinoIDE, GitHub

IOT & EDGE TECHNOLOGIES: Firebase, Supabase, Edge Impulse

PROFESSIONAL SKILLS: Problem Solving, Analytical Thinking, Time Management

INVOLVEMENTS

- Volunteered as part of the technical team for the college tech fest, supporting event coordination and technical arrangements. 2022-23 | 2024-45
- Part of the technical team for college Onam celebrations, assisting with sound, lighting, and event setup coordination | 2024

CERTIFICATION & PUBLICATION

- Research Publication: Smart Specs for Visually Impaired People using ESP32 – MAT Journals Pvt. Ltd | Vol. 2, Issue 2 | 2025
- Industry Immersion Program on Embedded Systems NeST Digital | June 2025
- Embedded Systems Internship (Arduino IDE) – Amal Jyothi College of Engineering (Autonomous)

PROJECTS

EMBEDDED / IoT PROJECTS:

BRAIN CONTROLLED WHEEL CHAIR USING EEG SIGNALS AND LINE FOLLOWING

Arduino IDE | ESP32 | FireBase | IoT | EEG/EOG

- Developing a brain-controlled wheelchair using EEG signals and eye-movement inputs for assisted navigation.
- Implemented alpha-wave-based BCI logic for embedded control on ESP32 with real-time data handling using Firebase.

SMART SPECS FOR VISUALLY IMPAIRED PEOPLE

Arduino IDE | ESP32 CAM | Edge Impulse | IoT

- Designed a wearable smart glasses system to help visually impaired people detect obstacles.
- Used sensors and a speaker to alert the user about nearby objects in real time.
- Focused on making the design simple, low-cost, and easy to use for daily life.

HUMAN ELDERLY MONITORING SYSTEM

Arduino IDE | ESP32 | FireBase | IoT

- Developed a compact chest-wearable IoT device using ESP32 for real-time health monitoring and fall detection, integrating MAX30102, MPU6050, DHT11, IR sensor, GPS, and GSM with buzzer and LED alerts for emergency notification.

HOME SAFETY - IOT BASED GAS LEAKAGE DETECTION SYSTEM

Arduino IDE | Arduino | FireBase | IoT

- Developed an IoT-based home safety system using ESP8266 to detect gas leakage and monitor humidity, with real-time data updates to Firebase and alert mechanisms using LEDs and a buzzer, as part of an internship at NeST Digital.

HARDWARE PROJECTS:

CUSTOM PCB BOARD FOR BCI- BASED WHEELCHAIR (FINAL YEAR PROJECT)

KiCad | 2 layer

- Designed manufactured and tested a custom PCB for EEG-based wheelchair control using KiCad, integrating power regulation, sensor integration, signal conditioning, and motor control interfaces.

5V–3.3V REGULATOR PCB

KiCad | 2 layer

- Designed a 5V to 3.3V voltage regulator PCB with an LDO and decoupling for stable power delivery.

UART CONVERTER PCB:

KiCad | 2 layer

- Designed a USB-to-UART converter PCB for reliable serial communication and programming.

MOTOR DRIVER PCB:

KiCad | 2 layer

- Designed a motor driver PCB using DR8855 for bidirectional DC motor control in embedded systems.

CUSTOM ARDUINO UNO DEV BOARD (ONGOING):

KiCad | 2 layer

- Currently designing an Arduino Uno – compatible development board with complete schematic and PCB layout.

CUSTOM MINI DEVELOPEMENT BOARD USING ESP32 C3 (ONGOING)

KiCad | 4 layer

- Currently designing an ESP32 C3 – compatible development board with complete schematic and PCB layout.